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Parachute Reefing

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Introduction

A reefed parachute is one in which the canopy opening is constricted to reduce drag production. This is done by rigging a cord around the periphery to reduce canopy volume. When an increase in drag is needed, the cord is cut allowing full canopy deployment.

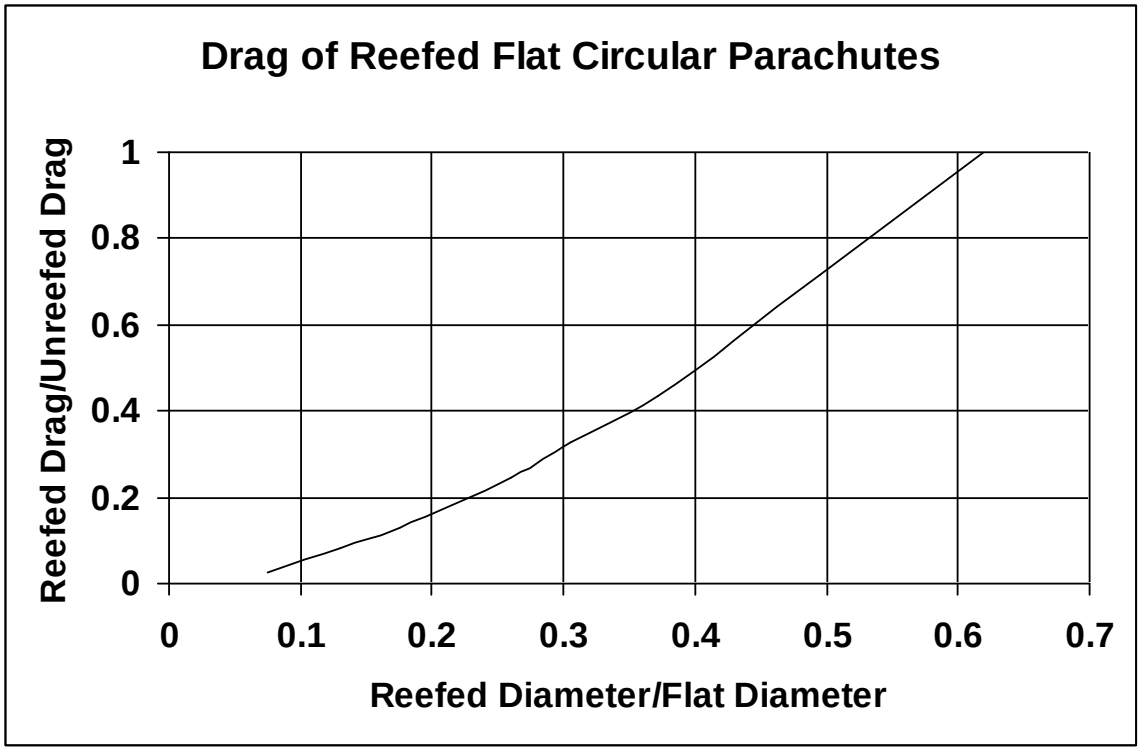
This memo answers the practical questions that arise with reefed parachute applications. For practical reasons, the data here is restricted to flat circular canopies with, or without, a vent hole at the center.

Nomenclature

Mnemonic	Definition
D_O	Canopy diameter when laid flat, ft
D_R	Diameter of the reefed canopy opening, ft
S_O	Area of the canopy when laid flat, neglecting vent hole area, if any, ft $=\frac{\pi}{4}D_O^2$
C_{DO}	Drag coefficient of the unreefed parachute, based on S_O
C_{DR}	Drag coefficient of the reefed parachute, based on S_O
C_S	Coefficient in the equation for reefing line strength
S	Breaking strength of the reefing line, lbs
q	Dynamic pressure just prior to parachute deployment, lbs/ft^2

Drag Reduction

The first question is how much drag reduction can be obtained. The experimental data taken from the reference is shown below. When unreefed, the canopy opening is about 62% of the diameter when the parachute is laid flat. The drag coefficient for a typical unreefed flat circular parachute is about 0.8 based on S_O.



Limitations on Reefing

Size matters when considering limits on reefing. For large parachutes ($D_O >$ 4m), the reefed diameter can be as small as 6% of D_O. But, for smaller parachutes, one would be wise the limit the reefed diameter to about 20% of D_O. Below these limits, parachutes can be unstable.

Reefing Line Attachment

While there are multiple answers to this question, the key issue is friction between the reefing line and its attachments to the canopy. With too much friction, full deployment when the reefing line is cut can be significantly impaired. The reference suggests that a metal reefing line ring be sewn to the canopy edge between each pair of suspension lines. Thus, if there are 6 suspension lines, the reefing cord would pass through 6 rings.

Multiple Reefs

It is feasible to implement more than one reefing line for a parachute to achieve staged increases in parachute drag. Just be careful to avoid tangling.

Reefing Line Strength

The required strength of a peripheral reefing cord is another common issue. Dimensional analysis suggests that the required strength be given by

$$S \geq q\left(\pi/4\right)D_O^2 C_S .$$

First, note that when $D_R = 0.62 D_O$ there is no reefing, and therefore, no tension load on any reefing cord. Assuming that C_S increases monotonically as D_R decreases, then data in the reference for a 28 ft personnel parachute indicate that

$$C_S \cong 0.02$$

should provide adequate strength for $D_R \geq 0.06 D_O$.

Disreefing

The process of restoring a reefed parachute to its full, unrestricted diameter is called disreefing. It is often accomplished with a piston-type guillotine cutter. The line to be cut passes transversely through the cylinder, and a black powder charge drives a piston down the cylinder where the line is sheared in two places. The only trick here is that ordinary nylon parachute cord is tough to shear. Attaching the reefing cord to a solid nylon strip used for forming wires into bundles will solve this problem.

Reference

T. W. Knacke, "Reefing of Parachutes", U. S. Air Force Aeronautical Systems Division Technical Report AD-A036 497, July 1976